

BarCo Corporation

Supply Chain Optimisation

Under Natural Gas Curtailment

A Linear Programming Approach to Production & Distribution Planning

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Presentation Overview

4 key sections — aligned with assessment criteria

Business Problem

01

Natural gas curtailment context
federal priority rules & financial risk

Analysis Approach

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Linear Programming formulation
model assumptions & methodology

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Production plan
distribution plan
sensitivity analysis & CO₂ scenario

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04

Three evidence-based strategic recommendations for management

The Business Problem

BarCo Corporation — Natural Gas Curtailment Crisis

Company Profile

- 8 chemical product lines across 3 plants (A, B, C)
- Gas supplier: MaxEnergy (nodes I, J, K)
- Current usage: $85,680 \times 10^3$ cu.ft/day
- Ammonia: gas as raw material (2nd Priority)
- All others: gas as boiler fuel (3rd Priority)

The Crisis

- ! MaxEnergy warns of natural gas shortages
- ! Extreme heat wave — power plants at full capacity
- ! Rolling brownouts: 20–40% curtailment expected
- ! Based on actual usage (85,680) not contract max
- ! MaxEnergy will NOT specify which products to cut
- ! BarCo has flexibility to absorb cuts strategically

Federal Commission Priority Rules

1st Priority

Residential & commercial heating/cooling

2nd Priority

Industrial raw material users ← BarCo Ammonia

3rd Priority

Industrial boiler fuel users ← BarCo (all other products)

Financial Risk at a Glance

Potential daily profit impact across curtailment scenarios

\$611,062

Baseline Daily Profit

\$513,023

20% Cut (–16.0%)

\$411,623

40% Cut (–32.6%)

\$358,650

50% Cut (–41.3%)

Key Insight:

Profit drops LESS than proportionally to gas — a 50% gas cut causes only 41.3% profit drop.

This is the value of LP-based smart prioritisation.

Value of LP-Driven Prioritisation vs. Naive Cut

- ✓ At 50% curtailment: LP saves ~\$44,000/day vs. a simple proportional cut
- ✓ Annualised value: ~\$16 million per year
- ✓ LP identifies exactly which products to protect — the optimal answer, not a guess

Analysis Approach: Linear Programming

The natural mathematical formalism for this problem

Why Linear Programming?

- Linear objective: maximise total profit (sum of profit $\times$ volume)
- Linear constraints: gas budget, daily capacity limits
- Continuous decision variables: production in tons/day
- Proven global optimal guarantee via Simplex algorithm
- Transparent, auditable, repeatable planning decisions
- Sensitivity analysis quantifies the impact of uncertainty

Model 1: Production LP

Maximise $Z = \sum_i p_i \cdot x_i$

Subject to:

$$\sum_i g_i \cdot x_i \leq G \quad (\text{gas budget})$$

$$0 \leq x_i \leq \text{cap}_i \quad (\text{capacity})$$

Model 2: Distribution LP

Minimise $T = \sum_i \sum_j c_{ij} \cdot y_{ij}$

Subject to:

$$\sum_j y_{ij} \leq s_i \quad (\text{supply node } i)$$

$$\sum_i y_{ij} = d_j \quad (\text{demand plant } j)$$

$$y_{ij} \geq 0$$

Key Model Assumptions

Conditions under which the LP solution is valid

Linearity

Profit/ton, gas/ton, and cost are constant — no economies of scale or diminishing returns.

Divisibility

Production volumes are continuous real numbers; rounding at ton-scale is trivial.

Certainty

Profit, gas, and cost coefficients known with certainty. Sensitivity analysis relaxes this.

Independent Caps

Each product operates independently up to its effective daily capacity ceiling.

Uniform Cut

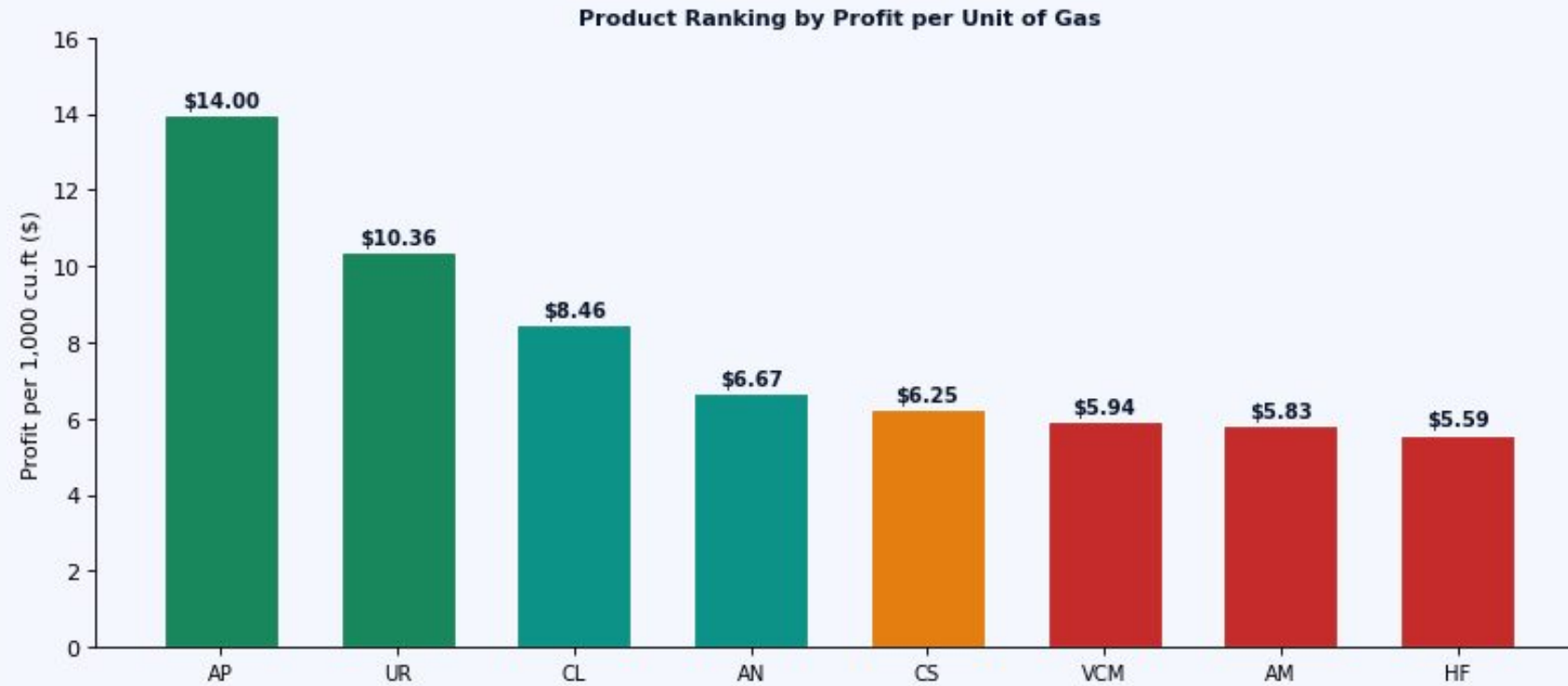
Gas reduction applies equally to all three supply nodes (I, J, K) per MaxEnergy.

Single-Period

LP optimises a representative single day; multi-period dynamics are a limitation.

Product Economics: LP Prioritisation Logic

Profit per unit of gas — ranking determines the optimal cut sequence



The Key Insight

Always protect:

- ① AP \$14.00 ② Urea \$10.36
③ Chlorine \$8.46 ④ AN \$6.67

Cut in this order:

- ⑧ HF \$5.59 (first cut)
⑦ AM \$5.83
⑥ VCM \$5.94
⑤ CS \$6.25 (last cut)

Not a heuristic —

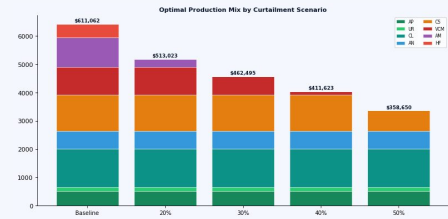
this IS the exact LP

solution!

Production Plan Results

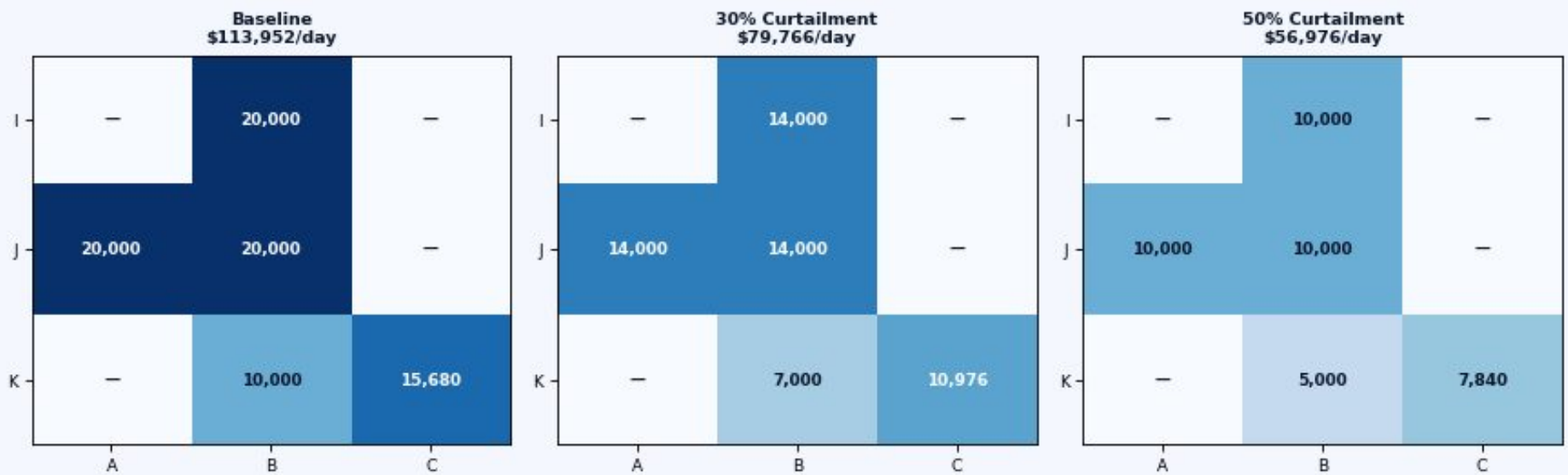
Optimal product mix under each curtailment scenario (LP-derived)

Product	Baseline	20% Cut	30% Cut	40% Cut	50% Cut
Ammonium Phosphate	510	510	510	510	510
Urea	150	150	150	150	150
Chlorine	1,350	1,350	1,350	1,350	1,350
Ammonium Nitrate	630	630	630	630	630
Caustic Soda	1,280	1,280	1,280	1,280	720
Vinyl Chloride Mon.	980	980	651	116	0
Ammonia	1,050	275	0	0	0
Hydrofluoric Acid	461	0	0	0	0
✓ LP protects 4 high-margin products at full capacity even at 50% cut.					
Cut sequence: HF → Ammonia → VCM → Caustic Soda					
vs. Baseline	—	-16.0%	-24.3%	-32.6%	-41.3%



Distribution Plan Results

Optimal gas allocation: MaxEnergy nodes (I,J,K) → BarCo plants (A,B,C)



Transport Costs (\$/1,000 cu.ft)			
	Plant A	Plant B	Plant C
Src I	\$1.30	\$1.20✓	\$1.60
Src J	\$1.20✓	\$1.40	\$1.50
Src K	\$1.80	\$1.60	\$1.40✓

- 5 active lanes: I→B, J→A, J→B, K→B, K→C | Baseline total cost: \$113,952/day
- Structure INVARIANT under curtailment — only volumes shrink | Avg unit cost: \$1.33 always
- Unused lanes I→C (\$1.60) and K→A (\$1.80) — renegotiate for better optionality

Sensitivity Analysis

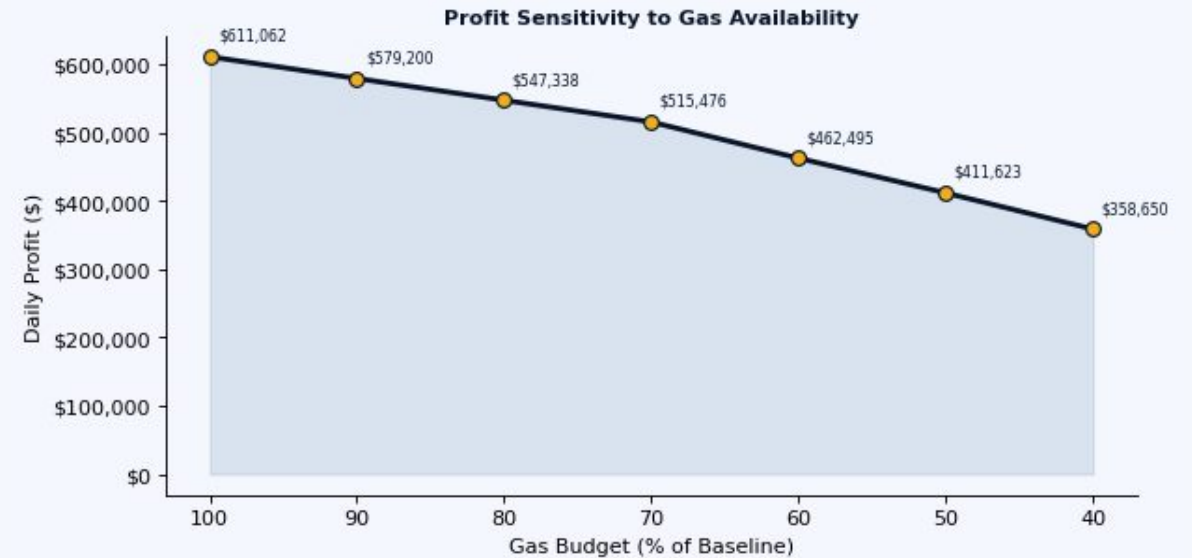
Shadow price of gas & profit robustness under coefficient changes

Shadow Price of Gas Constraint

Scenario	Gas Budget	Marginal Product	Shadow Price
Baseline	85,680	Hydrofluoric Acid	\$5.59/10 ³
20% Cut	68,544	Ammonia	\$5.83/10 ³
30% Cut	59,976	Vinyl Chloride Mon.	\$5.94/10 ³
40% Cut	51,408	Vinyl Chloride Mon.	\$5.94/10 ³
50% Cut	42,840	Caustic Soda	\$6.25/10 ³

Managerial Implication:

Every 1,000 cu.ft of spot gas secured above curtailed budget is worth \$5.59–\$6.25 in daily profit — defines the maximum price BarCo should pay for emergency gas procurement.

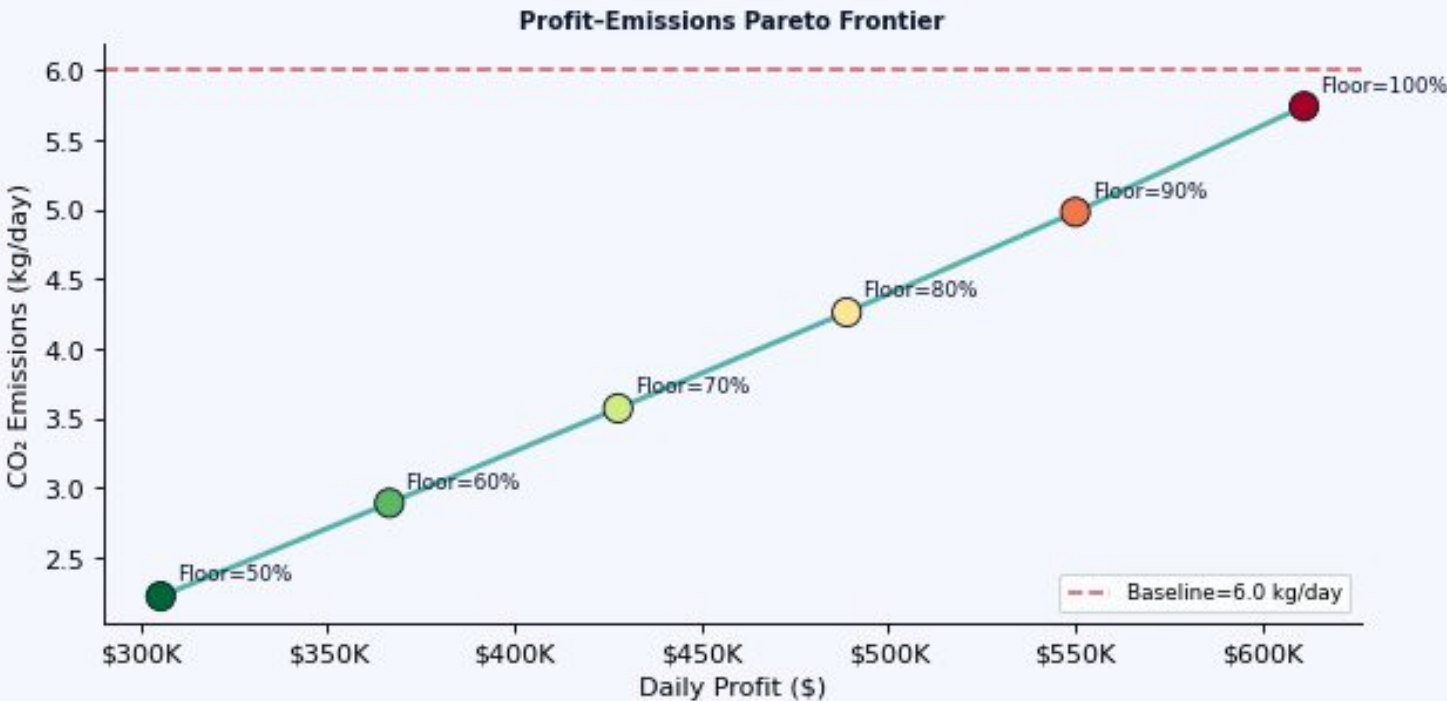


Profit Coefficient Stability

- Protected 4 products: stable within $\pm 20\%$ margin change — optimal mix unchanged
- Marginal products (HF, AM, VCM, CS): -20% margin removes them from optimal mix
- Rule: $<10\% \rightarrow$ no action | $10\text{--}20\% \rightarrow$ re-run LP monthly | $>20\% \rightarrow$ re-optimize immediately

Environmental Scenario: Minimising CO₂ Emissions

What if BarCo pursues sustainability alongside profit?



Profit–Emissions Trade-off

Profit Floor	Daily Profit	CO ₂ (kg)	CO ₂ Cut
100%	\$611,062	5.74 kg	4.4% ↓
90%	\$549,956	4.98 kg	17.0% ↓
80%	\$488,849	4.26 kg	29.0% ↓

Key Findings

- At profit-max, CO₂ already 4.4%
- below baseline (ammonia at full cap)
- 10% profit sacrifice → 17% CO₂ cut
- Break-even carbon: ~\$80,000/tonne
- AUS Safeguard today: ~\$30/tonne
- LP ready to activate when prices rise

Strategic Recommendations

Three evidence-based actions — each with quantified value

R1

Implement the LP-Derived Production Priority Rule

Always run AP, Urea, Chlorine, AN at full capacity. Cut sequence: HF → Ammonia → VCM → Caustic Soda.

Value: ~\$16M/year over naive cut | Shadow price: \$5.59–\$6.25/10³ cu.ft = max emergency gas price

R2

Commit to 5-Lane Distribution Structure

Pre-negotiate volume-flexible contracts: I→B, J→A, J→B, K→B, K→C. No need to re-tender each cycle.

Renegotiate unused I→C (\$1.60) and K→A (\$1.80) | Average cost locked at \$1.33 per 1,000 cu.ft

R3

Run CO₂-Aware LP in Parallel Each Cycle

Report Pareto trade-off to board each planning cycle. CO₂ already 4.4% below baseline at profit-max.

Activate CO₂-aware variant when AUS carbon prices clear ~\$80/tonne | Ahead of future regulation

Conclusion

LP transforms curtailment decisions

from reactive guesses into transparent, pre-committed policy

Profit drops sub-proportionally to gas

50% gas cut → 41.3% profit drop · LP saves ~\$16M/year vs naive approach

Shadow price quantifies gas value

\$5.59–\$6.25 per 1,000 cu.ft — exact ceiling for emergency procurement

Same model supports CO₂ planning

With these three recommendations, BarCo will not only weather the 2026 brownouts — it will build a planning capability fit for a more constrained, carbon-aware future.

Thank you · Questions welcome